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It Came from Another World!

Albert Einstein gave the last lecture of his life in Princeton, New Jersey, on April 14, 1954. It was a guest lecture in John Wheeler's graduate seminar on relativity, but the subject inevitably turned to the role of the observer in quantum physics. ("On quantum theory I use up more brain grease than on relativity," Einstein once told his friend Otto Stern.) Einstein outlined his objections to quantum physics; afterward, the students asked questions, attempting to defend Bohr's views as they had been taught by Wheeler. Einstein handled them with patience, asking in return, with a slight smile, "When a mouse observes, does that change the state of the universe?"

One of the first-year graduate students in the room that day made note of Einstein's pithy challenge to the Copenhagen interpretation. A year later, Einstein was dead, and the student, Hugh Everett, was putting Einstein's words to good use in defense of his own new interpretation of quantum physics. Unlike Einstein—and like Bohm—Everett attempted to resolve the problems of quantum physics through the mathematics of quantum physics itself, rather than attempting to find an entirely new theory. But, unlike Bohm, Everett's solution didn't involve pilot waves. Everett's answer to the questions that lurked at the foundations of quantum physics was wholly original—and far stranger than anything Bohm or Einstein had ever proposed.